The operator $\mathcal{A}$ is defined as

$$\mathcal{A} = -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\right)$$
$$D_{\mathcal{A}} = \{u \in \mathcal{C}2(-1,1) \mid u \text{ and } u' \text{ bounded/limited}\}$$

(a) Determine a scalar product in which $\mathcal{A}$ is symmetric.

(b) Show that $\mathcal{A}$ is symmetric in the scalar product from (a), using the definition.

(c) Show that $1, x, 2x^2 - 1$ are eigenfunctions to $\mathcal{A}$ and determine the corresponding eigenvalues.

(d) Evaluate

$$\int_{-1}^{1} \frac{2x^2-1}{\sqrt{1-x^2}}\,\mathrm{d}x$$

Solution:

(a) The operator is a Sturm-Liouville operator

$$\mathcal{A}u = -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\right) = -\frac{1}{w}(pu')', \qquad w(x) = \frac{1}{\sqrt{1-x^2}}$$

and the scalar product is then

$$(u \mid v) = \int_{-1}^{1} \overline{u(x)}v(x)\frac{1}{w(x)}\,\mathrm{d}x$$

(b) Using the scalar product definition

$$(u \mid v) = \int \overline{u(x)}v(x)w(x)\,\mathrm{d}x, \qquad w(x) = \frac{1}{\sqrt{1-x^2}}$$

we get

$$(u \mid \mathcal{A}v) =$$

$$= \int_{-1}^{1} \overline{u(x)}[\mathcal{A}v(x)]\frac{1}{w(x)}\,\mathrm{d}x = \int_{-1}^{1} \overline{u(x)}\left[-\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}v(x)\right)\right]\frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x =$$

$$= -\int_{-1}^{1} \overline{u(x)}\left(\sqrt{1-x^2}v'(x)\right)'\,\mathrm{d}x = -\underbrace{\left[\overline{u(x)}\sqrt{1-x^2}v'(x)\right]_{-1}^{1}}_{=0} + \int_{-1}^{1} \overline{u'(x)}\sqrt{1-x^2}v'(x)\,\mathrm{d}x =$$

$$= \int_{-1}^{1} \left(\overline{u'(x)}\sqrt{1-x^2}\right)v'(x)\,\mathrm{d}x = \underbrace{\left[\left(\overline{u'(x)}\sqrt{1-x^2}\right)v(x)\right]_{-1}^{1}}_{=0} - \int_{-1}^{1} \left(\overline{u'(x)}\sqrt{1-x^2}\right)'v(x)\,\mathrm{d}x =$$

$$= \int_{-1}^{1} -\left(\overline{u'(x)}\sqrt{1-x^2}\right)'v(x)\,\mathrm{d}x = \int_{-1}^{1} -\sqrt{1-x^2}\left(\overline{u'(x)}\sqrt{1-x^2}\right)'v(x)\frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x =$$

$$= \int_{-1}^{1} \left[-\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\overline{u(x)}\right)\right]v(x)\frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x =$$

$$= \int_{-1}^{1} \left[\mathcal{A}\overline{u(x)}\right]v(x)\frac{1}{w(x)}\,\mathrm{d}x = \boxed{(\mathcal{A}u \mid v)}$$

(c)

$$\mathcal{A}u = \lambda u$$
$$u(x) = 1$$
$$\downarrow$$
$$\mathcal{A}\cdot 1 = \lambda_0 \cdot 1 \to -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\cdot 1\right) = \lambda_0 \cdot 1 \to 0 = \lambda_0 \cdot 1 \to \boxed{\lambda_0 = 0}$$

$$u(x) = x$$
$$\downarrow$$
$$\mathcal{A}x = \lambda_1 x \to -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}x\right) = \lambda_1 x \to -\sqrt{1-x^2}\frac{-2x}{2\sqrt{1-x^2}} = \lambda_1 x \to$$
$$\to x = \lambda_1 x \to \boxed{\lambda_1 = 1}$$

$$u(x) = 2x^2 - 1$$
$$\downarrow$$
$$\mathcal{A}(2x^2-1) = \lambda_2(2x^2-1) \to -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}(2x^2-1)\right) = \lambda_2(2x^2-1) \to$$
$$\to -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(4x\sqrt{1-x^2}\right) = -\sqrt{1-x^2}\left(4\sqrt{1-x^2} - \frac{8x^2}{2\sqrt{1-x^2}}\right) = -4 + 4x^2 + 4x^2 =$$
$$= 8x^2 - 4 = \lambda_2(2x^2-1) \to 4(2x^2-1) = \lambda_2(2x^2-1) \to \boxed{\lambda_2 = 4}$$

(d) The integral can be rewritten with the scalar product in (a) as

$$\int_{-1}^{1} \frac{2x^2 - 1}{\sqrt{1 - x^2}} \, \mathrm{d}x = (1 \mid 2x^2 - 1), \qquad w(x) = \frac{q}{\sqrt{1 - x^2}}$$

Since the functions 1 and $2x^2 - 1$ are eigenfunctions, the scalar product between them are pairwise orthogonal and therefore $= 0$.

$\square$